ADHEESH RAMANATH

STUDENT RESEARCHER / EMBEDDED SYSTEMS / ROBOTICS

Plano, Texas | adheeshk.ramanath@gmail.com | github.com/adheeshram | linkedin.com/in/adheesh-ramanath

EDUCATION

Plano West Senior High School

Plano, Texas

Senior · Graduating May 2027

TECHNICAL SKILLS

LANGUAGES

Python, C++, C, Java, TypeScript, JavaScript, SQL, MATLAB, VBA, Modelica

FRAMEWORKS

FastAPI, Next.js, React, React Native, Expo, Firebase, OpenCV, MediaPipe

ENGINEERING

Fusion 360, SysML v2, sysml2-compiler, FMI/FMUs, Arduino, I2C, UART

DEVELOPMENT

Git, GitHub, OpenAI API, ElevenLabs API

SELECTED RECOGNITION

- FTC team: 4th, Texas UIL 6A State Championship
- FTC team: Inspire Award, North Texas qualifier
- ElevenLabs track winner, HackSMU VII
- Dallas Regional Science and Engineering Fair qualifier

FOCUS

Electrical and computer engineering, especially embedded systems, robotics, and the intersection of physical systems and AI.

RESEARCH

Georgia Institute of Technology

Jun-Oct 2026

Student Researcher · Prof. Sriram Vishwanath's ECE team

- Led the robotics/Modelica pipeline translating natural-language descriptions into compiled, executable physical-system models.
- Developed an NL-to-DSL verification harness spanning normalized specifications, retrieval, multi-expert generation, compiler validation, FMU export, simulation, behavioral checks, specification alignment, repair, and reproducible evidence collection.
- Hardened FMI/runtime execution against initialization, numerical-stability, discontinuity, bounded-state, and incomplete-trace failures; built resumable parallel experiments with provenance, checkpoints, failure tracing, and contamination-resistant evaluation.
- Produced reproducible evidence archives, cryptographic hashes, failure traces, and documented mentor handoffs.

ENGINEERING

FTC Robotics

Aug 2025-May 2026

Mechanical & Systems Engineer

CAD-designed and 3D-printed intake/sorting mechanisms, tuned PID controllers, integrated subsystems, and repaired match-day failures.

MedSight

Spring 2026

Co-Creator / Full-Stack Developer

Built a smart-glasses prototype for structured observations, baselines, deviation alerts, and longitudinal reports with FastAPI, Next.js, Firebase, Expo, OpenAI, and ElevenLabs.

GestureCanvas

2025-2026

Independent Engineering Research · LASER

Built an Arduino Uno/MPU6050 wearable that maps hand motion to drawings, using Python sensor fusion and ML correction to address drift.

EXPERIENCE & LEADERSHIP

Shining Mindz

Software Development Intern · May-Aug 2025

Built and deployed a Python/VBA ETL pipeline converting PowerPoint data into validated Excel reports; reduced processing time by 96%.

iCode

Paid Programming & Robotics Instructor · Spring-Summer 2026

Taught Python, Java, and robotics to 125+ students ages 7-14; designed curricula for 15+ project-based camps.

Leadership

HornetHacks · Career Cartography · Habitat for Humanity Club

Co-founded an in-development MLH hackathon targeting 300+ students; interviewed 15+ leaders and reached 50K+ viewers; built a 60+ member service club and raised $2K+ for affordable housing.